

Name: _____ Date: _____

Period: _____

Primary Objective

By the end of this lesson, I will be able to:

1. **Graph an equation in two variables by finding solutions and identifying explicit function pieces.**
2. **Determine whether two variables co-vary positively or negatively by examining the sign of $\Delta y/\Delta x$.**
3. **Identify where the rate of change is zero or undefined and interpret this geometrically.**

Vocabulary & Key Concepts**Implicit equation:**.....
.....**An equation in two variables where neither variable is isolated; it may describe one or more functions.****Explicitly defined function:**.....
.....**A function of the form $y = f(x)$, where y is solved for in terms of x .****Co-variation:**.....
.....**The way two quantities change together; characterized by the sign of $\Delta y/\Delta x$.****Rate of change ($\Delta y/\Delta x$):**.....
.....**The ratio of the change in y to the change in x between two nearby points. Positive $\Rightarrow$ same direction; negative $\Rightarrow$ opposite directions.****Hook***"The equation $x^2 + y^2 = 25$ is not a function — but it describes a circle. Can we still talk about how x and y change together? Can we split it into functions?"*Is $x^2 + y^2 = 25$ a function? Why or why not?**No — it fails the vertical-line test. For most x values there are two y values.****Part 1 — Graphing an Implicit Equation****To graph an equation in two variables, substitute specific values for one variable and solve for the other, then plot the resulting ordered pairs.**

Example: $x^2 + y^2 = 25$

Complete the table. (Hint: substitute x and solve for y .)

x	y (positive)	y (negative)
-5	0	0
-3	4	-4
0	5	-5
3	4	-4
5	0	0

This equation describes **2** explicit function(s):

$$y = \sqrt{25 - x^2} \quad (\text{upper half}) \quad y = -\sqrt{25 - x^2} \quad (\text{lower half})$$

Solving for one variable in an implicit equation gives **an explicit function whose graph is part or all of the graph of the original equation.**

Part 2 — Co-variation

Definition: Co-variation describes **how two quantities change together, including whether they change in the same or opposite directions.**

Example: Points (3, 4) and (4, 3) are both on the upper half of $x^2 + y^2 = 25$.

$$\frac{\Delta y}{\Delta x} = \frac{3 - 4}{4 - 3} = -1$$

Is this ratio positive or negative? **Negative**

What does this mean about how x and y change together?

As x increases, y decreases — they change in opposite directions.

Rule:

- If $\Delta y / \Delta x$ is **positive**, the variables **change in the same direction (both increase or both decrease).**
- If $\Delta y / \Delta x$ is **negative**, the variables **change in opposite directions.**
- If $\Delta y / \Delta x = 0$, the function has **horizontal** behavior.
- If $\Delta y / \Delta x$ is undefined, the function has **vertical** behavior.

Part 3 — Second Example: $xy = 4$

Solve for y : $y = 4/x$

x	y
1	4
2	2
4	1

Compute $\Delta y / \Delta x$ from $x = 2$ to $x = 4$:

$$\frac{\Delta y}{\Delta x} = \frac{1 - 2}{4 - 2} = -1/2$$

As x increases, y **decreases**. Co-variation is **negative**.

Teacher Note

Part 1: Many students write only the positive square root when completing the table. Reinforce that $y^2 = k$ always yields two solutions ($\pm\sqrt{k}$) unless $k = 0$.

Part 2: The fraction $\Delta y/\Delta x = -1$ at these two particular points. Emphasize that this is an approximation of the local behavior, not an exact derivative. The sign is what matters for co-variation.

Part 3: For $xy = 4$, $y = 4/x$. Students can verify: y decreases as x increases (for $x > 0$). Connect to the idea that the product of the two variables is always 4 — a fixed relationship.